Day 1 Teacher Guide | Lesson 3-5: Zeros of Polynomial Functions

Companion to student packet. 60-minute block, 7:45–8:45. Print count: 3 blank sign-analysis worksheets per student.

Time	Section	Min	Teacher Moves	Target
7:45	**Notice & Wonder**	8	Students do Think-Pair-Share on Model & Discuss. Circulate during pair share. Listen for: “zero means height is 0,” “set each factor = 0.” Pull 2–3 pairs to share. Use their words to write a class definition of “zero” on the board. Post definition for ELL reference. Don’t over-formalize — student language first.	*Move by 7:53*
7:53	**Example 1**	12	Model ONE row of the sign chart under doc cam: “In $x < -3$, test $x = -5$. Is x negative? Is $x - 4$ negative? Is $x + 3$ negative?” Release students to finish remaining 3 rows + sketch on blank worksheet #1. Circulate. If stuck on sketch, point to Product column: “Positive or negative here? What does that mean for the graph?” Desmos check: students graph $f(x) = x(x-4)(x+3)$ and compare.	*Desmos done by 8:05*
8:05	**Skill Unlock: Quad. Factoring**	3	Partners factor x^2+x-6 and x^2-5x+4. Equity-stick one pair per problem. If most stuck: “What two numbers multiply to −6 and add to 1?” Do NOT reveal that (a) is the exact quadratic inside Try It 1a — let them discover it.	*Move by 8:08*
8:08	**Try It 1a**	10	Students factor $4x^3+4x^2-24x$, find zeros, sign chart, sketch. Common error: forgetting GCF (trying to factor x^3+x^2-6x). Check during circulation. Correct zeros: −3, 0, 2. If a student asks “Do I use 4x or x as a column?” reply: “The 4 is always positive — does it change the sign?” Desmos check. Push: “What did the 4 do to the graph’s steepness?”	*Desmos done by 8:18*
8:18	**Skill Unlock: Diff. of Squares**	3	Partners factor x^2-4, x^2-81, then tackle x^4-81. Let pairs struggle with (c) for 90 sec. Then: “What if x^4 is a perfect square? What is it the square of?” Once someone says x^2, write $(x^2)^2 - 9^2$ on board. For “factor again?”: “Try diff. of squares on x^2+9. What goes wrong?” Seeds Day 4 complex zeros.	*Move by 8:21*
8:21	**Try It 1b**	12	Students factor x^4-81, find real zeros, sign	*Desmos done by*

			chart, sketch. Sign chart is the crux: 3 factor columns but x^2+9 is always positive. Ask: “If x^2+9 never changes sign, does it flip the product?” Push early finishers: “Why does this graph go up on both ends, unlike the cubics?” Desmos check.	*8:33*
8:33	**Compare**	4	Cold-call one pair for Q1 or Q3 from the Compare box. Target answer for Q3: “Desmos shows THAT the graph is above or below. The sign chart explains WHY.” Name that distinction aloud. This is your synthesis/discourse moment for admin observation.	*Move by 8:37*
8:37	**CER Exit Ticket**	8	Silent independent write. No partner talk. Circles back to Model & Discuss — students now have more context for why 2 zeros aren’t enough. Early finisher push: “If I gave you max height = 100 ft at t = 2.5, could you find the exact equation? Why?” Collect at 8:45.	*Collect 8:45*

Pressure Valves

Behind at 8:18? Cut the Diff. of Squares unlock to 2 min: skip parts (a)–(b), go straight to (c) x^4-81.

Behind at 8:21? Trim Try It 1b Desmos check to 1 min. Students verify zeros only, skip shape comparison.

Behind at 8:33? Pose Compare Q3 verbally, take 2–3 responses, skip written answers. Go straight to exit ticket.

Way behind at 8:35? Exit ticket becomes homework. Verbally close with: “We used the same 4-step process three times today. Tomorrow we look at what happens when zeros repeat.”

Quick Answer Key

Skill Unlock 1	(a) $(x+3)(x-2)$ (b) $(x-4)(x-1)$
Try It 1a Factor	$4x(x+3)(x-2)$ Zeros: −3, 0, 2
Try It 1a Signs	$x<-3$: $(-)(-)(-) = -$ | $-3<x<0$: $(-)(+)(-) = +$ | $0<x<2$: $(+)(+)(-) = -$ | $x>2$: $(+)(+)(+) = +$
Skill Unlock 2	(a) $(x+2)(x-2)$ (b) $(x+9)(x-9)$ (c) $(x^2+9)(x+3)(x-3)$ x^2+9 can’t factor: sum, not difference
Try It 1b Factor	$(x^2+9)(x+3)(x-3)$ Real zeros: −3, 3 only
Try It 1b Signs	$x<-3$: $(+)(-)(-) = +$ | $-3<x<3$: $(+)(+)(-) = -$ | $x>3$: $(+)(+)(+) = +$
CER Exit	Disagree. Two zeros give two points (0,0) and (5,0). Infinite parabolas pass through two points. Need a third point or the value of a to determine the unique equation.